

```
import java.util.Scanner;

public class Factorial {

    public static void main(String[] args) {
        // Create a scanner for user input
        Scanner scanner = new Scanner(System.in);

        // Prompt the user for a number
        System.out.println("Enter a number: ");
        int number = scanner.nextInt();

        // Check for negative numbers
        if (number < 0) {
            System.out.println("Factorial is not defined for negative numbers.");
            return;
        }

        // Calculate the factorial
        long factorial = calculateFactorial(number);

        // Check for overflow
        if (factorial == -1) {
            System.out.println("The factorial of " + number + " is too large to store in a long.");
        } else {
            System.out.println("The factorial of " + number + " is: " + factorial);
        }

        // Close the scanner
        scanner.close();
    }

    // Function to calculate the factorial of a number
    private static long calculateFactorial(int number) {
        long result = 1;
```

```
for (int i = 2; i <= number; i++) {  
    result *= i;  
  
    // Check for overflow  
    if (result < 0) {  
        return -1;  
    }  
}  
return result;  
}
```